

Portugal’s Public-Debt Interest Burden: Measurement, Dynamics, and Comparative Evidence

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Abstract

This report presents a reproducible empirical assessment of Portugal’s general-government interest burden. The object of study is interest payable by the general-government sector, measured in euros and as a percentage of nominal GDP. The main harmonised series is built from Eurostat ESA 2010 national accounts from 1995 onward; an AMECO linked extension is retained for pre-1995 historical context, with explicit accounting-basis labels. The analysis documents the long decline in the interest burden from the mid-1990s, the temporary reversal during the sovereign-debt crisis, the post-2015 compression of the effective interest rate, and the recent tension between higher market rates and a declining debt ratio. In 2025 Portugal paid EUR 5.96 billion in general-government interest, equal to 1.90 percent of GDP, while gross debt was 89.70 percent of GDP and the implicit interest rate was 2.18 percent. In the latest comparator panel Portugal ranked fourth of seven non-aggregate countries by interest expenditure as a percentage of GDP. The report combines descriptive statistics, accounting identities, debt-dynamics decomposition, cross-country comparison, and deterministic rate-shock simulations.

1 Introduction

Interest expenditure is a central fiscal variable because it links the legacy stock of public debt to current budgetary space. For a sovereign such as Portugal, the annual interest bill reflects past primary balances, nominal GDP growth, market conditions, debt-management choices, and the average maturity of the outstanding portfolio. A given euro amount of interest can represent very different fiscal burdens depending on nominal GDP and the public-debt ratio. Likewise, a market-rate increase does not immediately reprice the full public debt stock, because outstanding liabilities mature gradually.

This report studies the Portuguese general-government interest burden using a single reproducible pipeline. It deliberately separates three concepts that are often conflated in public debate: (i) interest payable under ESA national accounts, (ii) the interest burden as a ratio to GDP, and (iii) market yields on new or benchmark debt. The central dependent variable is the ratio of general-government interest payable to nominal GDP. The analysis also calculates an implicit interest rate on the debt stock, a primary-balance measure, a growth-deflator decomposition, debt-dynamics residuals, and counterfactual interest-rate sensitivities.

The report answers five substantive questions:

1. How much interest does Portugal’s general government pay each year?
2. How has the burden evolved relative to GDP?

3. Was the post-crisis decline driven by lower effective rates, a falling debt ratio, nominal growth, or inflation?
4. How quickly do rate shocks pass through under gradual refinancing?
5. How does Portugal compare with selected European peers under harmonised Eurostat definitions?

2 Data and Scope

2.1 Unit of observation

The unit of observation is a calendar year. The principal population is Portugal's general-government sector, ESA institutional sector **S13**. The main series begins in 1995 because that is the start of the harmonised ESA 2010 Eurostat series used by the pipeline. Earlier observations from AMECO are retained as a linked extension from 1960 to 1994. They are not allowed to overwrite Eurostat observations and are labelled as `linked_ESA2010_ESA95_ESA79`.

2.2 Source registry

The main Eurostat source tables are:

- `gov_10a_main`: general-government interest payable (D41PAY), overall balance (B9), and percentage-of-GDP counterparts.
- `gov_10dd_edpt1`: Maastricht general-government gross debt (GD) in million euros and percent of GDP.
- `nama_10_gdp`: nominal GDP (B1GQ, current-price million euros) and real GDP growth.
- `irt_lt_mcby_a`: long-term government bond yield.

The AMECO extension uses the all-CSV archive and the series codes configured in the project source registry. AMECO includes observed and forecast observations; forecast rows are labelled and excluded from historical descriptive statements unless explicitly noted.

2.3 Generated analytical files

The reproducible pipeline writes the following local outputs:

- `data/processed/portugal_debt_interest.csv`;
- `data/processed/portugal_debt_interest.sqlite`;
- `data/processed/eurostat_panel_metrics.csv`;
- `reports/validation.json`;
- `reports/summary.md`;
- publication-oriented figures in `reports/figures/`.

3 Measurement Framework

3.1 Interest burden

The main burden measure is:

$$b_t = \frac{I_t}{Y_t} \times 100, \quad (1)$$

where I_t is general-government interest payable and Y_t is nominal GDP. The processed dataset retains both the official Eurostat percentage-of-GDP series and a calculated ratio from million-euro interest and GDP. Reconciliation checks compare these two versions using configured tolerances.

3.2 Implicit interest rate

The default implicit interest rate is measured against average debt:

$$r_t = \frac{I_t}{(D_{t-1} + D_t)/2} \times 100, \quad (2)$$

where D_t is the year-end general-government debt stock. The average-debt denominator reduces distortion in years where the debt ratio moves sharply. A previous-year-debt denominator is available through configuration, but this report uses the default average-debt definition.

3.3 Primary balance

Interest expenditure is represented as a positive expense. The primary balance is therefore:

$$pb_t = ob_t + b_t, \quad (3)$$

where ob_t is the overall balance as a percent of GDP. This convention means that a country with an overall deficit can still run a primary surplus if the deficit is smaller than the interest bill.

3.4 Nominal growth, real growth, and GDP-deflator growth

Nominal GDP growth is calculated from current-price GDP. Real GDP growth is read from the Eurostat chain-linked volume series. GDP-deflator growth is derived multiplicatively:

$$1 + \pi_t^{GDP} = \frac{1 + g_t^{nominal}}{1 + g_t^{real}}. \quad (4)$$

3.5 Debt dynamics

The debt-dynamics identity is reported in percentage-point form:

$$\Delta d_t = \frac{r_t - g_t}{1 + g_t} d_{t-1} - pb_t + sfa_t. \quad (5)$$

The residual sfa_t is the stock-flow adjustment required to reconcile observed debt-ratio changes with the automatic debt-dynamics term and the primary balance.

4 Validation and Reproducibility

The dataset passed all error-severity validation checks in the latest local run. The validation file reports one warning: the calculated and official debt ratios differ by more than 0.15 percentage points in 1997 and 1998. This warning does not invalidate the dataset, but it is retained as an audit signal because the early linked series and source conventions can differ around accounting-basis transitions.

The latest local verification was:

- `pytest`: 133 tests passed;
- `ruff check .`: passed;
- `mypy src`: passed;
- the configured live pipeline completed successfully.

Generated raw, interim, processed, report, and figure artifacts are excluded from version control except for placeholder files. The analytical source code, configuration, tests, documentation, and this report are versioned.

5 Descriptive Findings

5.1 Long-run evolution

The observed dataset covers 1960–2025, with the harmonised ESA 2010 main series covering 1995–2025. The highest observed interest burden in the generated dataset occurred in 1995, at 5.50 percent of GDP. The lowest observed burden in the main recent period was 1.90 percent of GDP in 2022 and again in 2025. In 2025 the interest bill was EUR 5.96 billion, or 1.90 percent of GDP. The gross debt ratio was 89.70 percent of GDP, down from the pandemic peak of 134.10 percent in 2020.

Table 1: Selected annual observations

Year	Interest/GDP (%)	Interest (EUR m)	Debt/GDP (%)	Implicit rate (%)	Overall bal. (% GDP)	Primary bal. (% GDP)
1995	5.5	5038.9	62.2	–	-5.2	0.3
1996	4.8	4666.1	63.3	7.93	-4.7	0.1
2000	3.0	3861.9	54.2	5.69	-3.2	-0.2
2007	3.0	5205.1	72.7	4.16	-2.9	0.1
2011	4.3	7521.6	114.0	3.95	-7.7	-3.4
2014	4.8	8334.4	132.5	3.68	-7.4	-2.6
2019	2.9	6226.9	116.1	2.50	0.1	3.0
2020	2.8	5697.1	134.1	2.20	-5.8	-3.0
2021	2.4	5117.8	123.9	1.90	-2.8	-0.4
2022	1.9	4608.0	111.2	1.71	-0.3	1.6
2023	2.1	5553.3	96.9	2.08	1.1	3.2
2024	2.0	5934.8	93.5	2.23	0.6	2.6
2025	1.9	5964.5	89.7	2.18	0.7	2.6

5.2 Regime patterns

Table 2 summarises the ESA 2010 main series by descriptive regime. The sovereign-debt crisis and adjustment period is the highest-burden regime in the main series, with mean interest expenditure of 4.68 percent of GDP. The inflation and monetary-tightening regime has a much lower average interest burden, 1.98 percent of GDP, despite rising benchmark rates. The difference reflects the interaction of nominal GDP growth, a lower debt ratio, and the slow repricing of the debt stock.

Table 2: Regime summary, ESA 2010 main series

Regime	Start	End	Mean int./GDP	Max int./GDP	Mean debt/GDP
Euro convergence	1995	1998	4.30	5.50	59.95
Early euro period	1999	2007	2.82	3.00	64.07
Global financial crisis	2008	2010	3.00	3.10	87.67
Sovereign-debt crisis and adjustment	2011	2014	4.68	4.80	126.48
ECB intervention and refinancing	2015	2019	3.70	4.50	125.08
Pandemic	2020	2021	2.60	2.80	129.00
Inflation and monetary tightening	2022	2025	1.98	2.10	97.83

5.3 Recent observations

Recent dynamics are especially important because they show why the euro amount and the GDP ratio can move differently. From 2022 to 2025, the interest bill in million euros rose from EUR 4.61 billion to EUR 5.96 billion, but the burden remained around 2 percent of GDP. This is consistent with a declining debt ratio and strong nominal GDP growth offsetting part of the higher effective interest cost.

Table 3: Recent observed dynamics

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Implicit rate (%)	Nominal growth (%)	Real growth (%)
2018	3.3	6805.8	121.1	2.75	4.85	2.9
2019	2.9	6226.9	116.1	2.50	4.63	2.7
2020	2.8	5697.1	134.1	2.20	-6.27	-8.2
2021	2.4	5117.8	123.9	1.90	7.69	5.6
2022	1.9	4608.0	111.2	1.71	12.69	7.0
2023	2.1	5553.3	96.9	2.08	10.82	3.1
2024	2.0	5934.8	93.5	2.23	7.19	2.2
2025	1.9	5964.5	89.7	2.18	5.85	1.9

6 Figures

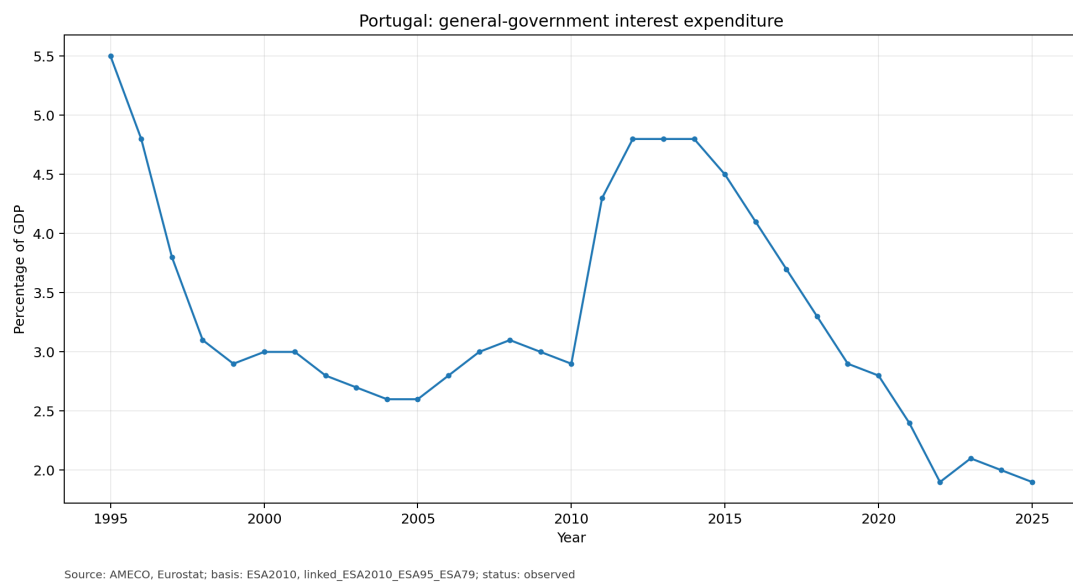


Figure 1: Interest expenditure as a percentage of GDP.

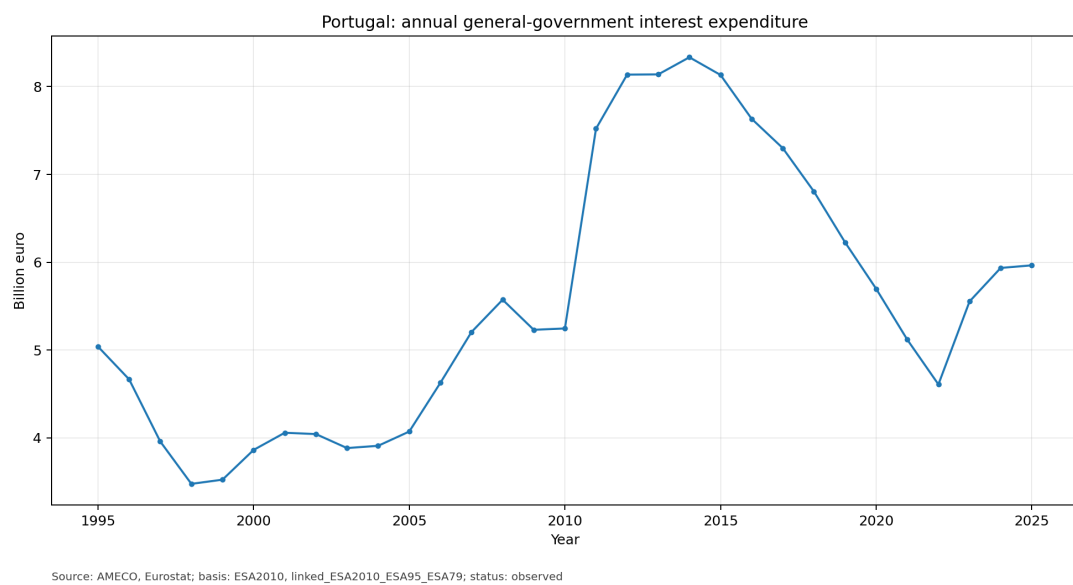


Figure 2: Interest expenditure in billion euros.

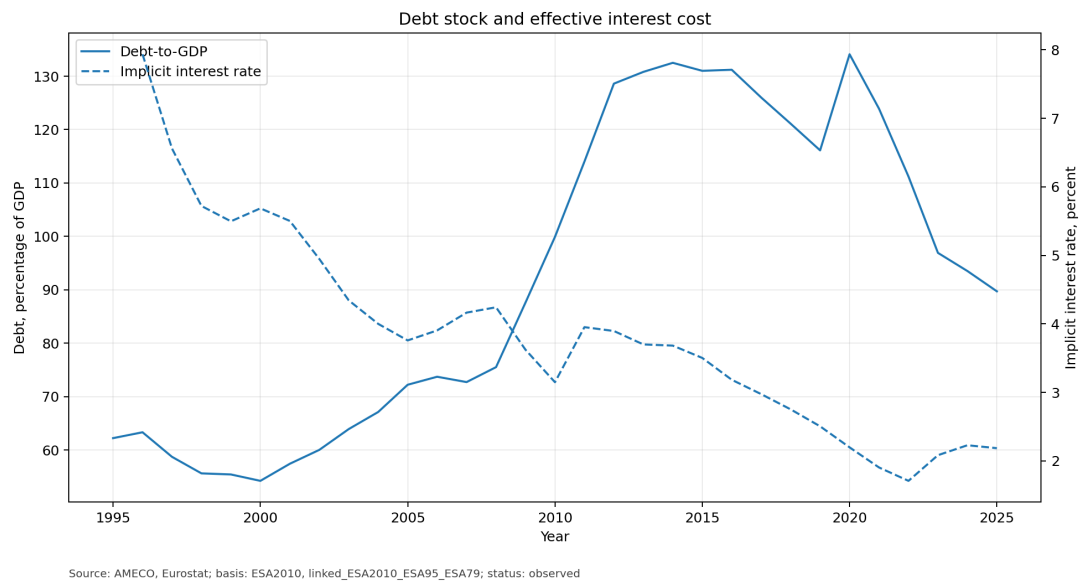


Figure 3: Debt ratio and implicit interest rate.

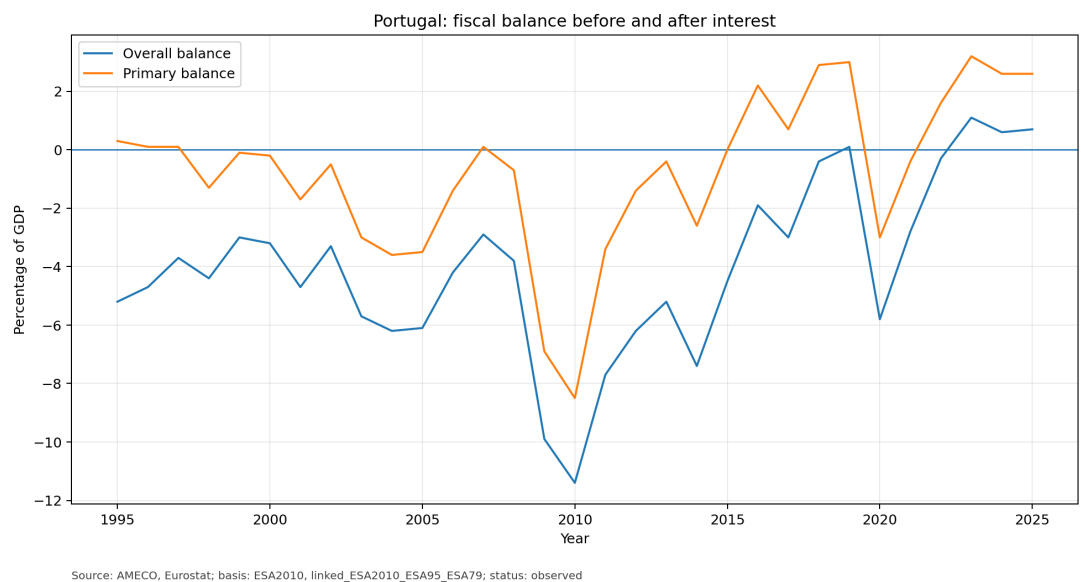


Figure 4: Overall and primary balances.

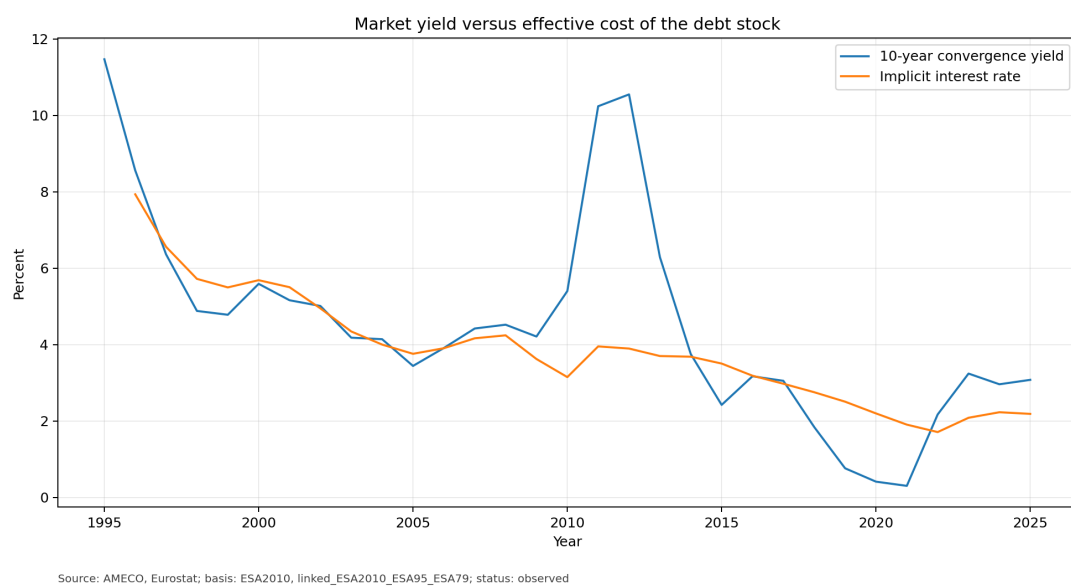


Figure 5: Benchmark market yield versus implicit interest rate.

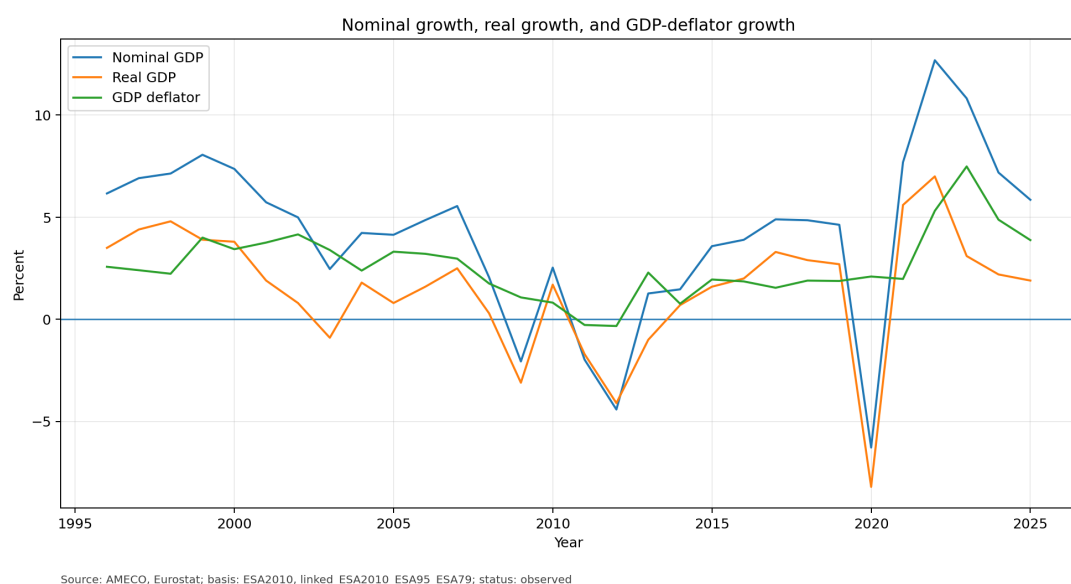


Figure 6: Nominal growth, real growth, and GDP-deflator growth.

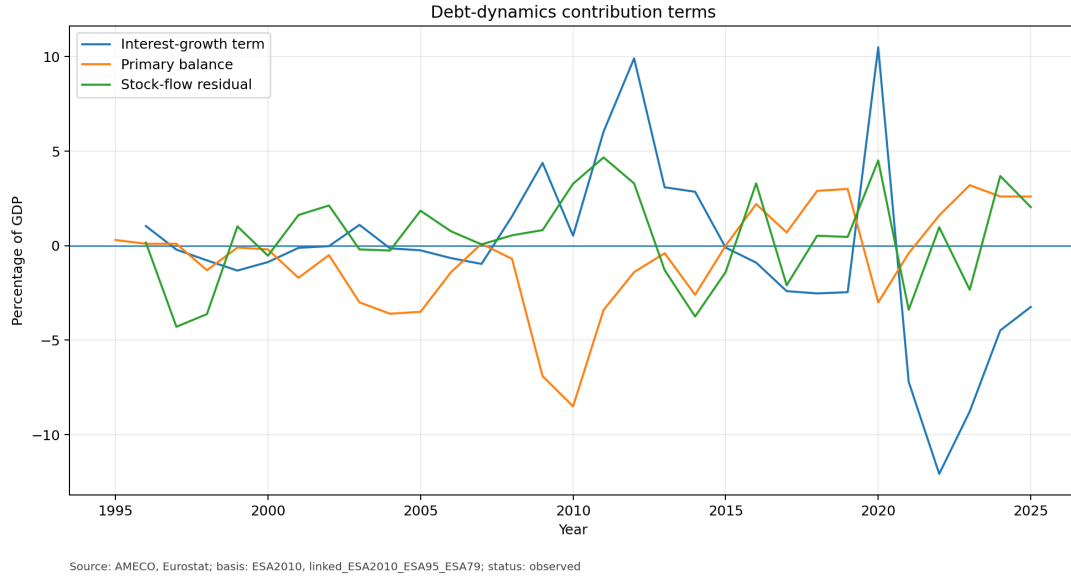


Figure 7: Debt-dynamics decomposition.

7 European Comparison

The comparator panel uses the same Eurostat concepts, units, and sector filters as the Portugal series. Aggregate geographies such as the euro area are retained for context but excluded from country ranks. In the 2025 panel Portugal ranked fourth of seven non-aggregate comparator countries by interest expenditure as a percentage of GDP.

Table 4: Comparator panel, latest observed year

Rank	Geography	Interest/GDP (%)	Debt/GDP (%)	Implicit rate (%)	Code
1	Italy	3.9	137.1	2.87	IT
2	Greece	3.2	146.1	2.15	EL
3	Spain	2.4	100.7	2.43	ES
4	Portugal	1.9	89.7	2.18	PT
5	Germany	1.1	63.5	1.79	DE
6	Netherlands	0.7	44.4	1.67	NL
7	Ireland	0.5	32.9	1.40	IE

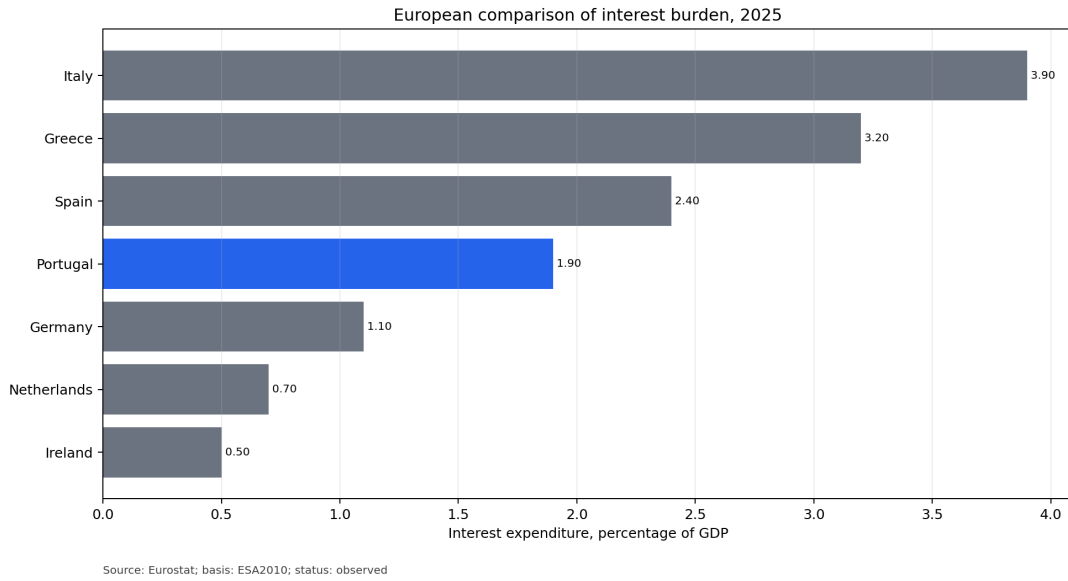


Figure 8: European comparison of interest expenditure as a percentage of GDP.

8 Rate-Shock Simulations

The static simulation applies a rate shock to the entire debt stock:

$$\Delta b = d\Delta r. \quad (6)$$

At the 2025 debt ratio of 89.70 percent of GDP, a full pass-through of 100 basis points implies an arithmetic increase in the interest burden of approximately 0.90 percentage points of GDP. The 50 and 200 basis-point shocks imply approximately 0.45 and 1.79 percentage points, respectively. These are long-run arithmetic effects, not one-year forecasts.

Table 5: Static full-pass-through sensitivities

Debt/GDP (%)	Shock (bps)	Shock rate decimal	Additional interest/GDP (pp)
89.70	50	0.005	0.449
89.70	100	0.010	0.897
89.70	200	0.020	1.794

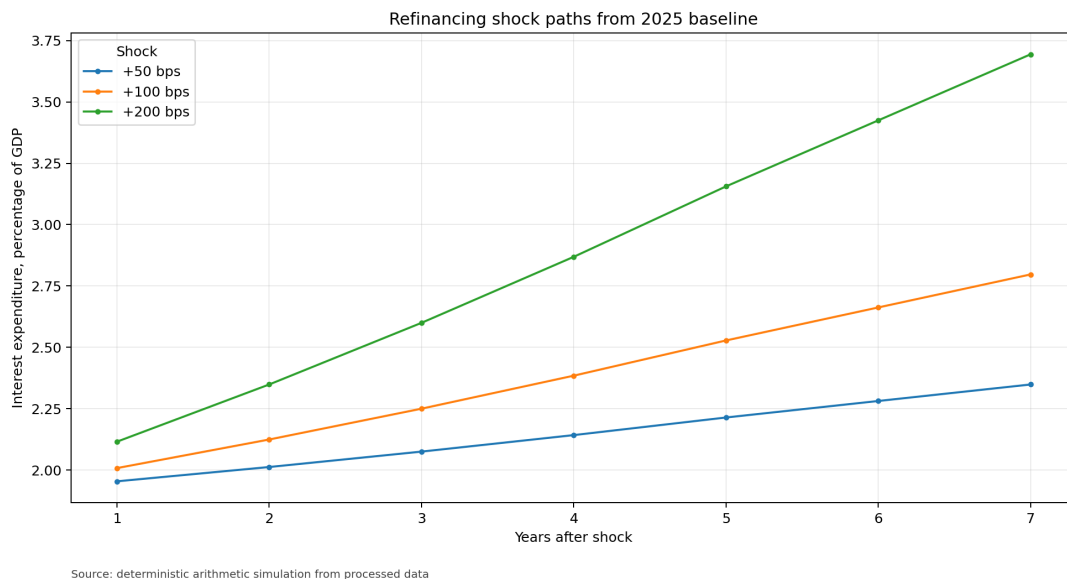


Figure 9: Gradual refinancing shock paths.

9 Interpretation

The central empirical finding is that Portugal's interest burden has fallen substantially from the mid-1990s and from the sovereign-debt-crisis peak. This decline should not be read as a simple decline in the euro amount of interest paid. Instead, it reflects several interacting mechanisms:

1. lower effective interest rates on the outstanding stock after the sovereign-debt crisis and the ECB intervention period;
2. strong nominal GDP growth in the post-pandemic and inflation period;
3. a declining debt ratio after the 2020 pandemic peak;
4. positive primary balances in recent years;
5. gradual rather than immediate pass-through from market yields to the average cost of the debt stock.

The European comparison places Portugal in an intermediate position. Its 2025 interest burden is below Italy, Greece, and Spain, but above Germany, the Netherlands, and Ireland. This ranking is consistent with Portugal's still-high but declining debt ratio and an implicit rate that remains moderate relative to the countries with the largest burdens.

10 Limitations

Several limitations should govern interpretation. First, pre-1995 observations are linked AMECO observations and do not have the same accounting status as the ESA 2010 Eurostat main series. Second, interest payable under ESA is not identical to central-government cash interest or to newly issued bond yields. Third, rate-shock simulations are deterministic arithmetic exercises; they do not model feedback from fiscal policy, maturity management, risk premia, inflation, or economic growth. Fourth, the stock-flow adjustment is a residual and should be interpreted as

a reconciliation term, not a structural causal estimate. Finally, cross-country comparisons are harmonised by Eurostat definitions but do not adjust for all country-level debt-management differences.

11 Conclusion

Portugal’s public-debt interest burden has moved from a high-burden environment in the mid-1990s and sovereign-debt-crisis period to a much lower burden in the current dataset. The 2025 interest bill remains economically material at EUR 5.96 billion, but the GDP burden is 1.90 percent, supported by a lower debt ratio and nominal growth. The main fiscal risk is not that market-rate increases immediately reprice the entire stock, but that sustained higher rates gradually raise the effective cost as debt is refinanced. The reproducible pipeline and validation checks in this repository make that distinction explicit and provide a basis for future updates as Eurostat and AMECO revise source data.

A Validation Appendix

The latest validation result passed all error-severity checks. One warning was recorded: official and calculated debt ratios differed by more than 0.15 percentage points in 1997 and 1998. This is retained as an audit item.

B Reproducibility Appendix

The report was prepared from the repository-local processed outputs generated by:

```
python -m pt_debt_interest.cli all --config config/default.yaml
```

The local quality checks were:

```
pytest
ruff check .
mypy src
```

C Source Notes

Eurostat datasets used by the pipeline include `gov_10a_main`, `gov_10dd_edpt1`, `nama_10_gdp`, and `irt_lt_mcby_a`. AMECO is used only for the linked historical extension and is preserved with explicit source and accounting-basis metadata.